

Magnetic Capsule Maze Cup

Steer a magnetic pill-camera through a body maze without touching it.

Win condition: Fastest clean run with low penalties and a clear magnetic-control explanation.

THE SCIENCE YOU ARE USING

Remote actuation. A magnet pulls a steel object through a barrier without contact. Doctors use the same idea to steer a swallowed capsule camera through the gut from outside the body. The field passes through the wall; only the magnetic force does the work.

Path planning. Move too fast and the capsule overshoots; move too slow and you waste time. You plan a route through the maze and control the external magnet to get a fast, clean run with the fewest wall touches.

HOW TO BUILD AND RUN IT

- 01 Test the field through a barrier.** Place the steel token on the maze and move a magnet under the board. Feel how the field pulls through.
- 02 Plan a route.** Trace the maze path you will follow and mark tricky corners.
- 03 Practice control.** Practice slow, steady moves. Sudden jerks make the token jump walls.
- 04 Run for time.** Make a timed clean run. Each wall touch is a penalty.
- 05 Explain the control.** Describe how distance and speed of the magnet changed your control.

Safety: You steer only the steel token (a paperclip or washer). NEVER touch or hold the tiny magnets. Swallowing two magnets, or a magnet plus a steel object, is a medical emergency. Keep all magnets away from mouths, electronics, and medical implants. A teacher counts magnets in and out. Allergy: None.

MATERIALS

Driver magnets sealed in bottle caps	controlled count
Maze boards	per team
Steel washers, nuts, or paper clips	per team
Timers	per team
Barrier sheets	shared

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): _____

The ONE variable we are testing: _____

Baseline result: _____

After redesign: _____

DEFEND YOUR DESIGN

What evidence made you change your design? _____

If you ran it again, what is the ONE thing you would change? _____

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Fastest clean run	50
Low penalties	20
Magnetic-control explanation	20
Redesign	10
Total	100